Proxy

Season: 2 · Structural · Category: GoF · Level: Beginner

Intent

- Surrogate controlling access to a real subject (lazy, remote, secure).

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Hotel key card opens only your floor.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- When a simpler redesign removes the need.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.structural.proxy;

interface Image {
    String display();
}

final class RealImage implements Image {
    private final String path;
    RealImage(String path) { this.path = path; }
    public String display() { return "showing:" + path; }
}

/** Virtual proxy — lazy-loads the real subject. */
public final class LazyImageProxy implements Image {
    private final String path;
    private RealImage real;

    public LazyImageProxy(String path) { this.path = path; }

    @Override
    public synchronized String display() {
        if (real == null) real = new RealImage(path);
        return real.display();
    }

    boolean loaded() { return real != null; }
}
```

Modern Java 25 / Functional

- Prefer sealed interfaces + records; virtual proxy uses synchronized lazy init.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/structural/proxy/
```

Pitfalls

- Keep adapters thin.
- Decorators should preserve the interface.
- Facades must not become god objects.

Interview Q&A

- Q: When is Proxy the wrong choice? A: When you can change the adaptee/subject API or collapse the hierarchy.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Surrogate controlling access to a real subject (lazy, remote, secure).
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Hotel key card opens only your floor. [0:40] Intent — State the design force in one sentence: Surrog...